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United States Patent	12415119
Kind Code	B2
Date of Patent	September 16, 2025
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Mixed material golf club head

Abstract

A hollow golf club head includes a metallic front body coupled with a composite rear body. The front body includes a strike face and a surrounding frame that extends rearward from a perimeter of the strikeface. The rear body includes a crown member coupled with a sole member. The sole member can have a structural layer formed from a filled thermoplastic material. The crown member can have structural layer and/or a resilient layer. The structural and resilient layers can be formed from fiber-reinforced thermoplastic composite material. The sole member has an opening through which a metallic weight pad at least partially extends. The weight pad is bonded to the structural layer and includes an aperture for attaching a metallic weight.

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Appl. No.: 18/311198

Filed: May 02, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230264079 A1	Aug. 24, 2023

Related U.S. Application Data

continuation parent-doc US 17249681 20210309 US 11638859 child-doc US 18311198
continuation parent-doc US 16714109 20191213 US 10940373 20210217 child-doc US 17249681
continuation parent-doc US 15901081 20180221 US 10300354 20190528 child-doc US 16380873

continuation parent-doc US 15607166 20170526 US 9925432 20180327 child-doc US 15901081
continuation-in-part parent-doc US 16380873 20190410 US 10765922 20200908 child-doc US
16714109
us-provisional-application US 62342741 20160527
us-provisional-application US 62779335 20181213

Publication Classification

Int. Cl.: **A63B53/04** (20150101); **A63B60/00** (20150101); **A63B60/02** (20150101)

U.S. Cl.:

CPC **A63B53/0475** (20130101); **A63B53/0466** (20130101); **A63B60/02** (20151001);
A63B53/04 (20130101); A63B53/0416 (20200801); A63B53/042 (20200801);
A63B53/0433 (20200801); A63B53/0437 (20200801); A63B53/045 (20200801);
A63B53/047 (20130101); A63B2053/0491 (20130101); A63B60/002 (20200801);
A63B2209/00 (20130101); A63B2209/02 (20130101)

Field of Classification Search

CPC: A63B (53/0475); A63B (60/02); A63B (53/0466); A63B (2209/02); A63B (2209/00);
A63B (53/047); A63B (53/04); A63B (2053/0491); A63B (53/042); A63B (53/045);
A63B (53/0433); A63B (53/0437); A63B (60/002)

USPC: 473/324-350

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This is a continuation of U.S. patent application Ser. No. 17/249,681, filed on Mar. 9, 2021, and is now U.S. Pat. No. 11,638,859, and which is a continuation of U.S. patent application Ser. No. 16/714,109, filed on Dec. 13, 2019, and is now U.S. Pat. No. 10,940,373, which claims the benefit of U.S. Provisional Appl. No. 62/779,335, filed on Dec. 13, 2018, and which is a continuation in part of U.S. patent application Ser. No. 16/380,873, filed on Apr. 10, 2019 and is now U.S. Pat. No. 10,765,922, which is a continuation of U.S. patent application Ser. No. 15/901,081, filed on Feb. 21, 2018 and is now U.S. Pat. No. 10,300,354, which is a continuation of U.S. patent application Ser. No. 15/607,166, filed on May, 26, 2017 and is now U.S. Pat. No. 9,925,432, which claims the benefit of U.S. Provisional Appl. No. 62/342,741, filed on May 27, 2016, the contents of all of which are incorporated fully herein by reference.

TECHNICAL FIELD

(1) This disclosure relates generally to a golf club head with a mixed material construction.

BACKGROUND

(2) In general, there are many important physical parameters (i.e., volume, mass, etc.) that effect the overall performance of a golf club head. One of the most important physical parameters, is the total mass of the golf club head. The total mass of the golf club head is the sum of the total structural mass and the total discretionary mass. Structural mass generally refers to the mass of the materials that are required to provide the club head with the structural resilience needed to withstand repeated impacts. Structural mass is highly design-dependent and provides a designer with a relatively low amount of control over specific mass distribution. Conversely, discretionary mass is any additional mass (beyond the minimum structural requirements of the golf club head) that may be added to the club head design for the sole purpose of customizing the performance and/or forgiveness of the club. There is a need in the art for alternative designs to all metal golf club heads to provide a means for maximizing discretionary weight to maximize club head moment of inertia (MOI) and lower/back center of gravity (CG).

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) This disclosure relates generally to sport equipment and relates more particularly to golf club heads and related methods.
- (2) FIG. 1 illustrates a bottom view of a mixed material golf club head.
- (3) FIG. 2 illustrates a top view of the golf club head of FIG. 1.
- (4) FIG. 3 illustrates a rear view of the golf club head of FIG. 1.
- (5) FIG. 4 illustrates an exploded view of the golf club head of FIG. 1.
- (6) FIG. 5 illustrates a front planar view of the golf club head of FIG. 1.
- (7) FIG. 6 illustrates rear planar view of a front body of the golf club head of FIG. 1.
- (8) FIG. 7 illustrates a rear view of the front body of the golf club head of FIG. 1.
- (9) FIG. 8 illustrates an exploded view of the front body and a rear body of the golf club head of FIG. 1.
- (10) FIG. 9 illustrates a cross sectional view of the golf club head of FIG. 1.
- (11) FIG. 10 illustrates an enlarged view of a weight pad and a weight in the golf club head of FIG. 1.
- (12) FIG. 11 illustrates an assembly view of a weight, a fastener, and a washer in the golf club head of FIG. 1.
- (13) FIG. 12 illustrates an internal view of the rear body of the golf club head of FIG. 1.
- (14) FIG. 13 illustrates an alternate internal view of the rear body of the golf club head of FIG. 1.
- (15) FIG. 14 illustrates is a schematic flow chart illustrating a method of manufacturing of the golf club head of FIG. 1.
- (16) Other aspects of the disclosure will become apparent by consideration of the detailed description and accompanying drawings.

DESCRIPTION

(17) Described herein is a golf club head that comprises a mixed material rear body in combination with a metallic front body, comprising a strike face and surrounding frame. The mixed material rear body is comprised of a fiber reinforced thermoplastic composite resilient layer, a molded thermoplastic structural layer, a metallic weight pad, and a metallic weight secured within the metallic weight pad. The mixed material rear body construction provides a significant reduction in structural mass, allowing for improved allocation of discretionary mass, thus improvements in the MOI and CG of the golf club head.

(18) The terms “first,” “second,” “third,” “fourth,” and the like in the description and in the claims, if any, are used for distinguishing between similar elements and not necessarily for describing a

particular sequential or chronological order. It is to be understood that the terms so used are interchangeable under appropriate circumstances such that the embodiments described herein are, for example, capable of operation in sequences other than those illustrated or otherwise described herein. Furthermore, the terms “include,” and “have,” and any variations thereof, are intended to cover a non-exclusive inclusion, such that a process, method, system, article, device, or apparatus that comprises a list of elements is not necessarily limited to those elements but may include other elements not expressly listed or inherent to such process, method, system, article, device, or apparatus.

(19) The terms “left,” “right,” “front,” “rear,” “top,” “bottom,” “over,” “under,” and the like in the description and in the claims, if any, are used for descriptive purposes and not necessarily for describing permanent relative positions. It is to be understood that the terms so used are interchangeable under appropriate circumstances such that the embodiments of the apparatus, methods, and/or articles of manufacture described herein are, for example, capable of operation in other orientations than those illustrated or otherwise described herein.

(20) Before any embodiments of the disclosure are explained in detail, it is to be understood that the disclosure is not limited in its application to the details of construction and the arrangement of components set forth in the following description or illustrated in the following drawings. The disclosure is capable of other embodiments and of being practiced or of being carried out in various ways.

(21) Described herein are various embodiments of a golf head having a mixed material construction. The mixed material construction comprises a metallic front body and a mixed material rear body. One embodiment of the club head includes a composite rear body with a metallic weight pad. In these or other embodiments, the rear body of the club head can include a fiber reinforced thermoplastic composite resilient layer, a molded thermoplastic structural layer, and a metallic weight secured within the metallic weight pad. In another embodiment, the rear body of the club head can include a composite crown and sole, with a metallic weight secured within the metallic weight pad. In many embodiments, the golf club head can be wood-type golf club head (i.e. driver, fairway wood, hybrid).

(22) In some embodiments, the club head can comprise a driver. In these embodiments, the loft angle of the club head can be less than approximately 16 degrees, less than approximately 15 degrees, less than approximately 14 degrees, less than approximately 13 degrees, less than approximately 12 degrees, less than approximately 11 degrees, or less than approximately 10 degrees. Further, in these embodiments, the volume of the club head can be greater than approximately 400 cc, greater than approximately 425 cc, greater than approximately 450 cc, greater than approximately 475 cc, greater than approximately 500 cc, greater than approximately 525 cc, greater than approximately 550 cc, greater than approximately 575 cc, greater than approximately 600 cc, greater than approximately 625 cc, greater than approximately 650 cc, greater than approximately 675 cc, or greater than approximately 700 cc. In some embodiments, the volume of the club head can be approximately 400 cc-600 cc, 425 cc-500 cc, approximately 500 cc-600 cc, approximately 500 cc-650 cc, approximately 550 cc-700 cc, approximately 600 cc-650 cc, approximately 600 cc-700 cc, or approximately 600 cc-800 cc.

(23) In some embodiments, the club head can comprise a fairway wood. In these embodiments, the loft angle of the club head can be less than approximately 35 degrees, less than approximately 34 degrees, less than approximately 33 degrees, less than approximately 32 degrees, less than approximately 31 degrees, or less than approximately 30 degrees. Further, in these embodiments, the loft angle of the club head can be greater than approximately 12 degrees, greater than approximately 13 degrees, greater than approximately 14 degrees, greater than approximately 15 degrees, greater than approximately 16 degrees, greater than approximately 17 degrees, greater than approximately 18 degrees, greater than approximately 19 degrees, or greater than approximately 20 degrees. For example, in some embodiments, the loft angle of the club head can

be between 12 degrees and 35 degrees, between 15 degrees and 35 degrees, between 20 degrees and 35 degrees, or between 12 degrees and 30 degrees.

(24) In embodiments where the club head comprises a fairway wood, the volume of the club head is less than approximately 400 cc, less than approximately 375 cc, less than approximately 350 cc, less than approximately 325 cc, less than approximately 300 cc, less than approximately 275 cc, less than approximately 250 cc, less than approximately 225 cc, or less than approximately 200 cc. In these embodiments, the volume of the club head can be approximately 150 cc-200 cc, approximately 150 cc-250 cc, approximately 150 cc-300 cc, approximately 150 cc-350 cc, approximately 150 cc-400 cc, approximately 300 cc-400 cc, approximately 325 cc-400 cc, approximately 350 cc-400 cc, approximately 250 cc-400 cc, approximately 250 cc-350 cc, or approximately 275 cc-375 cc.

(25) In some embodiments, the club head can comprise a hybrid. In these embodiments, the loft angle of the club head can be less than approximately 40 degrees, less than approximately 39 degrees, less than approximately 38 degrees, less than approximately 37 degrees, less than approximately 36 degrees, less than approximately 35 degrees, less than approximately 34 degrees, less than approximately 33 degrees, less than approximately 32 degrees, less than approximately 31 degrees, or less than approximately 30 degrees. Further, in these embodiments, the loft angle of the club head can be greater than approximately 16 degrees, greater than approximately 17 degrees, greater than approximately 18 degrees, greater than approximately 19 degrees, greater than approximately 20 degrees, greater than approximately 21 degrees, greater than approximately 22 degrees, greater than approximately 23 degrees, greater than approximately 24 degrees, or greater than approximately 25 degrees.

(26) In embodiments where the club head comprises a hybrid, the volume of the club head is less than approximately 200 cc, less than approximately 175 cc, less than approximately 150 cc, less than approximately 125 cc, less than approximately 100 cc, or less than approximately 75 cc. In some embodiments, the volume of the club head can be approximately 100 cc-150 cc, approximately 75 cc-150 cc, approximately 100 cc-125 cc, or approximately 75 cc-125 cc.

(27) FIG. 1-10 illustrate an embodiment of a golf club head **100** having a metallic front body **104**, and a rear body **108**. The front body **104** and rear body **108** are secured together to define a substantially closed/hollow interior volume. As is conventional with wood-style golf heads, the golf club head **100** includes a crown **112**, a sole **116**, and can be divided into a heel region **124** and a toe region **128**.

(28) In some embodiments, the golf club head **100** comprises a metallic front body **104**, and a composite rear body **108**, wherein the rear body comprises a woven fiber reinforced thermoplastic resilient layer **148**, a molded thermoplastic structural layer **152**, and a metallic weight pad **156**. The combination of a woven fiber reinforced thermoplastic resilient layer **148** and a molded thermoplastic structural layer **152**, enables savings in structural mass, in comparison to a similar club head made entirely from metal.

(29) The structural weight savings achieved by using a resilient layer **148** and a structural layer **152**, can be used to either reduce the entire weight of the club head **100** (which may provide faster club head speed and/or long hitting distances) or to increase the amount of discretionary mass that is available for placement on the golf club head **100**. In one embodiment, the additional discretionary mass, gained from using a composite resilient layer **148** and a composite structural layer **152**, can be reintroduced into the club head **100** in the form of a metallic weight pad **156**. The combination of a light composite rear body **108** and metallic weight pad **156**, allow the club head **100**, to allocate a mass majority of the club head in a position to maximize the MOI and CG, leading to more forgiveness and longer shots.

(30) I. Front Body

(31) Referring to FIGS. 4-7, the front body **104** of the club head **100** comprises a strike face **120**, intended to impact a golf ball. The front body **104** comprises a surrounding frame **136** that extends

rearward from a perimeter **140** of the strike face **120**, to provide the front body **104** with a cup-shaped appearance. The surrounding frame **136** comprises an internal surface **170** and an external surface **172**. Furthermore, the surrounding frame **136** can comprise a flange **174**, to provide an attachment surface to connect the front body **104** and the rear body **108**. When the front body **104** is combined with the rear body **108**, the external surface **172** of the front body **104** forms a portion of the crown **112** and the sole **116** of the club head **100**. The front body **104** further comprises a hosel **144** for receiving a golf club shaft or shaft adapter in the heel region **124** of the golf club head **100**.

(32) In some embodiments, the strike face **120** and surrounding frame **136** can be integrally formed. In other embodiments, the strike face **120** and surrounding frame **136** can be separately formed and joined together. In one embodiment, the strike face **120** is forged and the surrounding frame **136** is cast, then the strike face **120** and surrounding frame **136** are joined through welding, brazing, plasma welding, low-power laser welding, forging, or another suitable joining technique.

(33) In many embodiments, the front body **104** is made from a metallic material to withstand the repeated impact stress from striking a golf ball. In some embodiments, the front body **104**, can be formed from stainless steel, titanium, aluminum, a steel alloy (e.g. 455 steel, 475 steel, 431 steel, 17-4 stainless steel, maraging steel), a titanium alloy (e.g. Ti 7-4, Ti 6-4, T-9S), an aluminum alloy, or a composite material. In some embodiments, the strike face **120** of the golf club head **100** can comprise stainless steel, titanium, aluminum, a steel alloy (e.g. 455 steel, 475 steel, 431 steel, 17-4 stainless steel, maraging steel), a titanium alloy (e.g. Ti 7-4, Ti 6-4, T-9S), an aluminum alloy, an amorphous metal alloy, or a composite material.

(34) The front body **104** comprises a mass. In some embodiments, wherein the strike face **120** and surrounding frame **136** are separate, the mass of the front body **104** is the sum of the mass of the strike face **120** and the mass of the surrounding frame **136**. Depending on the material the front body **104** is made of, the mass of the front body **104** can range between 40 grams and 140 grams. In most embodiments, the mass of the front body **104** does not exceed 140 grams. In some embodiments, the mass of the front body **104** can range between 40-50 grams, 50-60 grams, 60-70 grams, 70-80 grams, 80-90 grams, 90-100 grams, 100-110 grams, 110-120 grams, 120-130 grams, or 130 grams-140 grams.

(35) a. Strike Face

(36) Referring to FIGS. 5, 6, and 9, the front body **104** of the golf club head **100** comprises a strike face **120**, positioned to strike a golf ball. The strike face **120** comprises a centerpoint **160**, a loft plane **164**, and a midplane **168**. The center point **160** is equidistant from the crown **112** and sole **116** of the club head **100**, and equidistant from the edge of the face that is the most proximate to the toe region **128** and from the edge of the strike face **120** that is the most proximate to the heel region **124**. The loft plane **164** is tangent to the centerpoint **160** of the strike face **120** of the club head **100**. The loft plane **164** intersects a ground plane **180**.

(37) The strike face **120** of the club head **100** comprises a thickness measured as the distance between the strike face **120** and the internal surface **170** of the front body **104**. The thickness of the strike face **120** varies at different locations defining a variable face thickness (VFT) or variable thickness profile **196**. The variable thickness profile **196** having a central region **192** and a peripheral region **188**. In many embodiments, the central region **192** of the variable thickness profile **196** comprises an ellipse or oval or ovoid or egg-like shape. The central region **192** is generally oblong and extends from a portion of the strike face **120** near the sole **116** and heel region **124** to a portion of the strike face **120** near the toe region **128** and crown **112**.

(38) Referring to FIG. 6, the central region **192** extends over or is positioned on or near the centerpoint **160** of the strike face **120** such that the center point **160** of the strike face **120** is located in the central region **192**. The central region **192** comprises a maximum thickness of the strike face **120**. In many embodiments, the thickness of the central region **192** is substantially constant. The peripheral region **188** is positioned around the perimeter **140** of the strike face **120** and comprises a

minimum thickness of the strike face **120**. In many embodiments, the thickness of the peripheral region **188** is substantially constant. The thickness of the strike face **120** in the central region **192** is greater than the thickness of the strike face **120** in the peripheral region **188**. A transition region **190** is positioned between the central region **192** and the peripheral region **188**. The transient region **190** includes a varying thickness that creates a transition between the central region **192** and the peripheral region **188**.

(39) Furthermore, the strike face **120** comprises a major axis **184** extending in a general heel **124** to toe **128** direction. The major axis **184** intersects the centerpoint **160** and forms an angle β with the ground plane. In many embodiments, the major axis **184** reflects the oblong shape of the central region **192**.

(40) The major axis **184** forms an approximate angle of 20 degrees with the ground plane **180**. For example, the angle formed between the major axis **184** of the central region **192** and the ground plane **180** can vary from 0 to 60 degrees. In some embodiments, the angle formed between the major axis **184** of the central region **192** and the ground plane **180** can vary from 2 to 20, 2 to 30, 5 to 40, 10 to 50, or 15 to 60 degrees. In other embodiments, the major axis **184** can create an angle of 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, or 60 degrees with ground plane **180**. By disposing the center region **192** on an angle it further allows the elongated portion of the egg-shape to extend towards the upper-toe portion of the strike face **120** wherein high CT values exist, thus improving resulting ball speed.

(41) The oval or ovoid or egg-like shape, along with the angle of the central region **192** of the variable thickness profile **196**, enables thicker regions of the strike face **120** to be positioned in regions having inherently high CT, and thinner regions of the strike face **120** to be positioned in regions having inherently low CT. Accordingly, regions of the face having inherently high CT are reduced, and regions of the face having inherently low CT are increased, resulting in normalized CT across the strike face **120**. In many embodiments, the variable thickness profile **196** results in a range in characteristic time less than 115 micro-seconds (μ s), less than 110 μ s, less than 105 μ s, less than 100 μ s, less than 95 μ s, less than 90 μ s, or less than 85 μ s. Further, in many embodiments, the variable thickness profile **40** results in an average characteristic time greater than 230 μ s, greater than 235 μ s, or greater than 240 μ s. For example, in many embodiments, the average CT of the face plate **20** can be between 230 μ s and 240 μ s, between 235 μ s and 240 μ s, or between 240 μ s and 245 μ s.

(42) Further, because the angled VFT is designed to position thickened portions of the strike face **120** in regions where it is required, the strike face **120** can experience a weight reduction compared to a strike face devoid of the variable thickness profile **196** described herein. The extra discretionary weight can be re-introduced in other regions of the club head to manipulate the club head center of gravity position and to increase club head moment of inertia, further improving the performance of the club head. In the illustrated embodiment, the club head **100** having the variable thickness profile **196**, as described herein, saves 2.1 grams of weight compared to a similar club head devoid of the variable thickness profile **196**.

(43) b. Hosel

(44) The front body **104** of the golf club head **100** comprises the hosel **144**. The hosel **144** includes a hosel axis **176** extending along a center of a bore of the hosel **144**. Referring to FIGS. 3 and 6, in the present example, a hosel coupling mechanism of the golf club head **100** comprises the hosel **144** and a shaft sleeve (not shown), where the shaft sleeve can be coupled to an end of a golf shaft (not shown). The shaft sleeve can couple with the hosel **144** in a plurality of configurations, thereby permitting the golf shaft to be secured to the hosel **144** at a plurality of angles relative to the hosel axis **176**. There can be other examples, however, where the shaft can be non-adjustably secured to the hosel **144**. In the illustrated embodiment, the hosel axis **176** is at an angle α with the ground plane **12** with respect to a front view of the golf club head **10** (FIG. 1). The illustrated angle α is

approximately 60-degrees, but in other constructions, the angle α may be between approximately 40-80 degrees (e.g., approximately 40 degrees, approximately 45 degrees, approximately 50 degrees, approximately 55 degrees, approximately 60 degrees, approximately 65 degrees, approximately 70 degrees, approximately 75 degrees, or approximately 80 degrees).

(45) Furthermore, the hosel axis **176** and the major axis **184** form an angle θ . In many embodiments, the angle θ formed between the hosel axis **176** and the major axis **184** can range between 60 and 140 degrees. In most embodiments, the minimum angle θ formed between the hosel axis **176** and the major axis **184** is approximately 60 degrees. In some embodiments, the angle θ formed between the hosel axis **176** and the major axis **184** can range between 60-70 degrees, 70-80 degrees, 80-90 degrees, 90-100 degrees, 100-110 degrees, 110-120 degrees, 120 degrees-130 degrees, or 130-140 degrees. In one embodiment, the angle the angle θ formed between the hosel axis **176** and the major axis **184** can range between 80 degrees and 90 degrees.

(46) c. Surrounding Frame

(47) The front body **104** of the golf club head **100** comprises the surrounding frame **136** that extends rearward from the entire perimeter **140** of the strike face **120**. The surrounding frame **136** further comprises a flange **174** that is operative to couple the front body **104** and the rear body **108**.

(48) The flange **174** provides a surface, to achieve a lap joint, wherein the rear body **108** can attach. The flange **174** extends rearward from the entire surrounding frame **136**, and forms a step-type structure, down from the external surface **172** of the surrounding frame **136**. In many embodiments, the flange **174** of the front body **104** allows the rear body to overlap the flange **174** and join to the front body **104**, by way of epoxy, adhesion, welding, bonding, laser assisted metal-plastic welding, brazing, or any other suitable attachment method. The lap joint style flange **174**, further allows the front body **104** and rear body **108** to securely mate, without the use of any mechanical fasteners.

(49) Furthermore, the surrounding frame **136** comprises the external surface **172** and the internal surface **170**, wherein additional aerodynamic features can be placed, to improve the overall speed of the golf club head. The surrounding frame **136** of the front body **104** of the golf club head **100**, can include additional aerodynamic features, such as turbulators **200**. The turbulators **200** can be used to reduce club head drag and increase the speed of the club **100**. These turbulators **200** are further described in U.S. Pat. No. 9,555,294, which is incorporated by reference in its entirety.

(50) II. Rear Body

(51) Referring to FIGS. **4**, and **8-13**, the rear body **108** of the club head **100** comprises a crown member **204**, a sole member **208**, and a weight pad **212**. The crown member **204** and sole member **208** are bonded together to form a portion of the crown **112** and the sole **116** of the golf club head **100**. When the front body **104** and rear body **108** are joined, the external surface **172** of the front body **104**, the crown member **204**, and the sole member **208**, form the entire crown **112** and sole **116** of the golf club head **100**. The sole member **208** of the rear body **108** can further comprise a composite resilient layer **152**, a composite structural layer **156**, and a metallic weight pad **212**.

(52) In the present design, the rear body **108** may include a mix of molded thermoplastic materials (e.g., injection molded thermoplastic materials) and fiber reinforced thermoplastic composite materials. As used herein, a molded thermoplastic material is one that relies on the polymer itself to provide structure and rigidity to the final component. The molded thermoplastic material is one that is readily adapted to molding techniques such as injection molding, whereby the material is freely flowable when in a heated to a temperature above the melting point of the polymer. A molded thermoplastic material with a mixed-in filler material is referred to as a filled thermoplastic (FT) material. Filled thermoplastic materials are freely flowable when in a heated/melted state. To facilitate the flowable characteristic, filler materials generally include discrete particulate having a maximum dimension of less than about 25 mm, or more commonly less than about 12 mm. For example, the filler materials can include discrete particulate having a maximum dimension of 4 mm, 5 mm, 6 mm, 7 mm, 8 mm, 9 mm, or 10 mm. Filler materials useful for the present designs may include, for example, glass beads or discontinuous reinforcing fibers formed from carbon,

glass, or an aramid polymer.

(53) In contrast to molded and filled thermoplastic materials, fiber reinforced composite (FRC) materials generally include one or more layers of a uni- or multi-directional fiber fabric that extend across a larger portion of the polymer. Unlike the reinforcing fibers that may be used in FT materials, the maximum dimension of fibers used in FRCs may be substantially larger/longer than those used in FT materials and may have sufficient size and characteristics such that they may be provided as a continuous fabric separate from the polymer. When formed with a thermoplastic polymer, even if the polymer is freely flowable when melted, the included continuous fibers are generally not.

(54) FRC materials are generally formed by arranging the fiber into a desired arrangement, and then impregnating the fiber material with a sufficient amount of a polymeric material to provide rigidity. In this manner, while FT materials may have a resin content of greater than about 45% by volume or more preferably greater than about 55% by volume, FRC materials desirably have a resin content of less than about 45% by volume, or more preferably less than about 35% by volume. FRC materials traditionally use two-part thermoset epoxies as the polymeric matrix, however, it is possible to also use thermoplastic polymers as the matrix. In many instances, FRC materials are pre-prepared prior to final manufacturing, and such intermediate material is often referred to as a prepreg. When a thermoset polymer is used, the prepreg is partially cured in intermediate form, and final curing occurs once the prepreg is formed into the final shape. When a thermoplastic polymer is used, the prepreg may include a cooled thermoplastic matrix that can subsequently be heated and molded into final shape. This technique enables complex and lightweight geometries to be made, such as the rear body **108**, without sacrificing strength.

(55) a. Crown Member

(56) The rear body **108**, comprises the crown member **204**. Referring to FIGS. **4** and **9** the crown member **204** comprises an external surface **206**, such that when the rear body **108** and front body **104** are joined, the external surface **206** of the crown member **204** and the external surface **172** of the surrounding frame **136** form the entire crown **112** of the golf club head **100**. The external surface **206** of the crown member **204** comprises a generally curvilinear shape which is concave with respect to the ground plane **180**. The generally curvilinear shape of the crown member **204** allows the rear body **208** to seamlessly be joined to the front body **104**, as the crown member is placed entirely over the flange **174** of the front body **104**.

(57) In many embodiments, the crown member **204** is comprised of a carbon fiber weave, devoid of any layering of composite plies or unidirectional composite plies. In one embodiment, the crown member **204** may be substantially formed from a formed fiber reinforced composite material that comprises a woven glass or carbon fiber reinforcing layer embedded in a polymeric matrix. In such an embodiment, the polymeric matrix is preferably a thermoplastic material such as, for example, polyphenylene sulfide (PPS), polyether ether ketone (PEEK), or a polyamide such as PA6 or PA66. In other embodiments, the crown member **204** may instead be formed from a filled thermoplastic material that comprises a glass bead or discontinuous glass, carbon, or aramid polymer fiber filler embedded throughout a thermoplastic material such as, for example, polyphenylene sulfide (PPS), polyether ether ketone (PEEK), or polyamide. In still other embodiments, the crown member **204** may have a mixed-material construction that includes both a filled thermoplastic material and a formed fiber reinforced composite material.

(58) b. Sole Member

(59) The rear body **108**, comprises the sole member **208**. Referring to FIGS. **4** and **9**, the sole member **208** comprises the structural layer **156** and the resilient layer **152**, providing a lightweight, but strong sole **116** of the golf club head **100**. In reference to the ground plane **180**, the resilient layer **152** is positioned tangent to the ground plane, and the structural layer **156** is placed on top of the resilient layer **152**, in the interior of the golf club head **100**.

(60) In one embodiment, the sole member **208** has a mixed-material construction that includes both

a fiber reinforced thermoplastic composite resilient layer **152** and a molded thermoplastic structural layer **156**. In a preferred embodiment, the molded thermoplastic structural layer **156** may be formed from a filled thermoplastic material that comprises a glass bead or discontinuous glass, carbon, or aramid polymer fiber filler embedded throughout a thermoplastic material such as, for example, polyphenylene sulfide (PPS), polyether ether ketone (PEEK), or a polyamide such as PA6 or PA66. The resilient layer **152** may then comprise a woven glass, carbon fiber, or aramid polymer fiber reinforcing layer embedded in a thermoplastic polymeric matrix that includes, for example, a polyphenylene sulfide (PPS), a polyether ether ketone (PEEK), or a polyamide such as PA6 or PA66. In one particular embodiment, the crown member **202** and resilient layer **152** may each comprise a woven carbon fiber fabric embedded in a polyphenylene sulfide (PPS), and the structural layer **156** may comprise a filled polyphenylene sulfide (PPS) polymer.

(61) The structural layer **156** may generally include a forward portion **236** and a peripheral portion **240** that define an outer perimeter of the sole member **208**. In an assembled club head **100**, the forward portion **236** is bonded to the metallic front body **104**, and the peripheral portion **240** is bonded to the crown member **204**. The structural layer **156** defines a plurality of apertures **244** located interior to the perimeter that each extend through the thickness of the structural layer **156**. Further, the structural layer **156** may include one or more structural members **248** that extend from the forward portion **236** and between at least two of the plurality of apertures **244**. Furthermore, as described below, the structural layer **156** can be configured to comprise a metallic weight pad **212** and metallic weight **220**.

(62) The resilient layer **152** may be bonded to the structural layer **156** such that it directly abuts or overlaps at least a portion of the forward portion **236**, the peripheral portion **240**, and the plurality of structural members **248**. In doing so, the resilient layer **152** may entirely cover each of the plurality of apertures **244** when viewed from the exterior of the club head **100**. Likewise, the one or more structural members **248** may serve as selective reinforcement to an interior portion of the resilient layer **244**, akin to a reinforcing rib or gusset.

(63) With respect to both the polymeric construction of the crown member **204** and the sole member **208**, any filled thermoplastics or fiber reinforced thermoplastic composites should preferably incorporate one or more engineering polymers that have sufficiently high material strengths and/or strength/weight ratio properties to withstand typical use while providing a weight savings benefit to the design. Specifically, it is important for the materials of the golf club head **100** to efficiently withstand the stresses imparted during an impact between the strike face **120** and a golf ball, while not contributing substantially to the total weight of the golf club head **100**. In general, preferred polymers may be characterized by a tensile strength at yield of greater than about 60 MPa (neat), and, when filled, may have a tensile strength at yield of greater than about 110 MPa, or more preferably greater than about 180 MPa, and even more preferably greater than about 220 MPa. In some embodiments, suitable filled thermoplastic polymers may have a tensile strength at yield of from about 60 MPa to about 350 MPa. In some embodiments, these polymers may have a density in the range of from about 1.15 to about 2.02 in either a filled or unfilled state and may preferably have a melting temperature of greater than about 210° C. or more preferably greater than about 250° C.

(64) c. Weight Pad

(65) With reference to FIGS. **4** and **9-11**, in many embodiments, the structural layer **156** can include a weight pad **212**. The weight pad **212** comprises a cavity **216** adapted to receive a metallic weight **220**. In some embodiments, the weight pad **212** is generally located toward the rear most point on the club head **100**, and therefore may be integral to and/or directly coupled with the rear portion **132** of the structural layer **156**. In some embodiments, a hole or opening **252** may be provided in the resilient layer **152**, through which a portion of the weight pad **212** may extend. In some embodiments, the opening **250** is spaced apart from the front body **104** by a minimum distance of at least 25 mm, or at least 30 mm, or at least 35 mm (i.e., measured along the outer

surface of the club head). As shown in FIG. 9, when assembled, an outer surface of the weight pad **212** may sit flush with an outer surface of the directly adjacent sole member **208** and/or resilient layer **152**. In this manner, a portion of the weight pad **212** may form part of the external sole **116** of the golf club head **100**. Additionally, in some embodiments, an internal surface of the weight pad **212** may be exposed on an interior of the clubhead. The weight pad **212** functions to provide a dense rearward mass to improve the overall MOI of the golf club head. The weight pad **212** and weight **220** provide a region to place a high concentration of discretionary mass, since there are substantially weight savings achieved from forming the composite rear body **108**.

(66) The weight pad **212** can comprise any desired shape, in order to position as much mass towards the periphery of the rear portion **132** of the golf club head **100**. The shape of the weight pad **212** can be any one of the following shapes: circular, triangular, square, rectangular, trapezoidal, pentagonal, curvilinear, spade-shaped, or any other polygon or shape with at least one curved surface. In one embodiment, the weight pad **212** can be a roughly trapezoidal shape. In another embodiment, the weight pad **212** can be a roughly rectangular shape. Furthermore, in another embodiment, the weight pad **212** can be a roughly circular shape. Further still, in another embodiment, the weight pad **212** can be a roughly triangular shape.

(67) In most embodiments, the weight pad **212** can be made from a metallic material to provide a dense rearward portion to improve the overall MOI of the golf club head **100**. In some embodiments, the weight pad **212** can be formed from stainless steel, titanium, aluminum, a steel alloy (e.g. 455 steel, 475 steel, 431 steel, 17-4 stainless steel, maraging steel), a titanium alloy (e.g. Ti 7-4, Ti 6-4, T-9S), an aluminum alloy, or a composite material. In one embodiment, the weight pad **212** can be made from a stainless steel. The weight pad **212** can be forged or cast, prior to being secured within the sole member **208** of the rear body **108**.

(68) The weight pad **212** may be secured within the opening **250** in resilient layer **152** through via one or more techniques that are operable to provide a robust, structural bond. Due to differences in material types/material surface energies, as well as the comparatively high ratio of component mass to contact surface area, it may be difficult for conventional adhesives alone to withstand the forces experienced during a golf club impact with a ball. As such, it may be desirable to integrate at least a portion of the weight pad into the structural layer **156** and/or resilient layer **152** by encapsulating at least a portion of the weight pad. In doing so, the material strength of the encapsulating layer may be operative to provide a more durable bond than the use of surface adhesives alone. Referring to FIGS. 9 and 13, examples of suitable encapsulation may include structural tapes **261** extending over an edge **252** of the weight pad **212**, direct encapsulation of at least a portion of the weight pad **212** by the structural layer **156**, or encapsulation of a portion of the weight pad between adjacent plies of the resilient layer **152**. These techniques may be used instead of, or in addition to the use of chemical adhesives provided between the weight pad and sole member **208**.

(69) In one configuration, the weight pad **212** may be attached to the sole member **208** without the use of any mechanical fasteners. In one embodiment, the weight pad **212** is casted and then the structural layer **156** may be molded around at least the edge **252** of the weight pad **212**, for example, via an insert injection molding or comolding technique. As noted above, the filled thermoplastic construction of the structural layer **156** is particularly suited to receive the weight pad **212** due to its ability to form complex geometry and extend around edges in a structurally stable manner. Depending on the geometry of the weight pad, such a joining technique may be more difficult with tapes or FRCs due to their more uniform profile.

(70) The cavity **216** of the weight pad **212** extends inward from weight pad **212**. In the illustrated embodiment, the cavity **216** comprises a circular shape. In other embodiments, the cavity **216** can comprise any shape. For example, the shape of the cavity **216** can comprise a circle, an ellipse, a triangle, a rectangle, an octagon, or any other polygon or shape with at least one curved surface. The cavity **216** provides a recess to affix a metallic weight **220** within. The metallic weight **220**, further adds discretionary weight to the golf club head **100**, thus further improving the MOI and

CG of the golf club head **100**. Additionally, the cavity **216** and metallic weight **220** allow for changes to be made to the overall weight of the golf club head **100**, by removably attaching different metallic weights of different densities.

(71) The cavity **212** includes a depth measured from a base **224** of the cavity **212** to the external contour of the sole member **208**, in a direction generally perpendicular to the base **224**. In many embodiments, the depth of the cavity **212** is between 0.10 inches and 0.50 inches. In some embodiments, the depth of the cavity **212** is less than 0.50 inches, less than 0.45 inches, less than 0.40 inches, less than 0.35 inches, less than 0.30 inches, less than 0.25 inches, less than 0.20 inches, or less than 0.15 inches.

(72) Further, the cavity **212** comprises an aperture **228** in the base **224**. The aperture **228** extends inward from the base **224** of the cavity **212**, towards the crown **112** of the golf club head **100**. In some embodiments, the aperture **228** can comprise threading that mates with the threading of a fastener **230** to secure the metallic weight **220** within the cavity **216**. In other embodiments, the aperture **228** can be devoid of threading for use with a self-tapping or self-drilling fastener.

(73) The metallic weight **220** is configured to be positioned with the cavity **216** of the weight pad **212**. In the illustrated embodiment, the weight **220** is circular in shape to correspond to the shape of the cavity **212**. In other embodiments, the weight **220** can comprise any geometric shape corresponding to the shape of the cavity **212** (e.g., circular, elliptical, triangular, rectangular, trapezoidal, octagonal, or any other polygonal shape or shape with at least one curved surface).

(74) The metallic weight **220** further comprises an aperture **232** extending entirely through the weight **220**. The aperture **232** is substantially similar in size to the aperture **228** of the cavity **212** and the aperture **232** of the weight **220** aligns with the aperture **228** of the cavity **212**, when the weight is positioned within the cavity **212**. In most embodiments, the aperture **232** is devoid of threading to allow the fastener **230** to pass through the weight **220** and secure, via threading, to the aperture **228** of the weight pad **212**. Additionally, in some embodiments, a washer **214** can be positioned in the cavity **212** prior to the positioning of the metallic weight **220** within the cavity **212**.

(75) d. Assembly

(76) FIG. **14** illustrates an embodiment of a method **300** for manufacturing a golf club head **100** having the integrally bonded resilient layer **152**, structural layer **156**, and metallic weight pad **220** of the sole member **208**. The method **300** involves thermoforming a fiber reinforced thermoplastic composite into an external shell portion of the club head **100** at step **310**. The thermoforming process may involve, for example, pre-heating a thermoplastic prepreg to a molding temperature at least above the glass transition temperature of the thermoplastic polymer, molding the prepreg into the shape of the shell portion, and then trimming the molded part to size.

(77) Once the composite shell portion is in a proper shape, a filled polymeric supporting structure may then be injection molded into direct contact with the shell at step **320**. Such a process is generally referred to as insert-molding. In this process, the shell is directly placed within a heated mold having a gated cavity exposed to a portion of the shell. Molten polymer is forcibly injected into the cavity, and thereafter either directly mixes with molten polymer of the heated composite shell, or locally bonds with the softened shell. As the mold is cooled, the polymer of the composite shell and supporting structure harden together in a fused relationship. The bonding is enhanced if the polymer of the shell portion and the polymer of the supporting structure are compatible and is even further enhanced if the two components include a common thermoplastic resin component. While insert-molding is a preferred technique for forming the structure, other molding techniques, such as compression molding, may also be used.

(78) With continued reference to FIG. **14**, once the sole member **208** is formed through steps **310** and **320**, an FRC crown member **204** may be bonded to the sole member **208** to substantially complete the structure of the rear body **108** (step **330**). In a preferred embodiment, the crown member **204** may be formed from a thermoplastic FRC material that is formed into shape using a

similar thermoforming technique as described with respect to step 310. Forming the crown member 204 from a thermoplastic composite allows the crown member 204 to be bonded to the sole member 208 using a localized welding technique. Such welding techniques may include, for example, laser welding, ultrasonic welding, or potentially electrical resistance welding if the polymers are electrically conductive. If the crown member 204 is instead formed using a thermoset polymer, then the crown member 204 may be bonded to the sole member 208 using, for example, an adhesive or a mechanical affixment technique (studs, screws, posts, mechanical interference engagement, etc).

(79) The rear body 108, comprising the affixed crown member 204 and sole member 208 may subsequently be adhesively bonded to the metallic front body 104 at step 340. While adhesives readily bond to most metals, the process of adhering to the polymer may require the use of one or more adhesion promoters or surface treatments to enhance bonding between the adhesive and the polymer of the rear body 108.

(80) III. Benefits

(81) Utilizing a mixed material rear body construction can provide a significant reduction in structural weight while not sacrificing any design flexibility and providing a robust means for reintroducing discretionary mass. While such a design may be formed entirely from a filled thermoplastic, such as polyphenylene sulfide (PPS), as discussed above, the use of a fiber reinforced composite provides a stronger and lighter construction across continuous outer surfaces. Conversely, an all-FRC design would not readily incorporate weight-receiving structures, and thus would not be able to easily capitalize on increased discretionary mass.

(82) The metallic weight pad is beneficial over a mixed material golf club head devoid a metallic weight pad because the metallic weight pad allows for variance and interchangeability of the metallic weight, while providing a durable and secure location to affix the metallic weight. In comparison to a golf club head devoid of the metallic weight pad, the metallic weight pad securely withstands the torque imparted on the weight pad when a weight is being affixed. Further, the metallic weight pad allows for the manufacturer to interchange the metallic weight, to adjust for manufacturing tolerances (i.e., change the desired swing weight of the overall club head from 206 grams to 209 grams), or adjust for customer specification (i.e., a golfer wants his/her club head heavier, 206 grams to 209 grams).

(83) Replacement of one or more claimed elements constitutes reconstruction and not repair. Additionally, benefits, other advantages, and solutions to problems have been described with regard to specific embodiments. The benefits, advantages, solutions to problems, and any element or elements that may cause any benefit, advantage, or solution to occur or become more pronounced, however, are not to be construed as critical, required, or essential features or elements of any or all of the claims.

(84) As the rules to golf may change from time to time (e.g., new regulations may be adopted or old rules may be eliminated or modified by golf standard organizations and/or governing bodies such as the United States Golf Association (USGA), the Royal and Ancient Golf Club of St. Andrews (R&A), etc.), golf equipment related to the apparatus, methods, and articles of manufacture described herein may be conforming or non-conforming to the rules of golf at any particular time. Accordingly, golf equipment related to the apparatus, methods, and articles of manufacture described herein may be advertised, offered for sale, and/or sold as conforming or non-conforming golf equipment. The apparatus, methods, and articles of manufacture described herein are not limited in this regard.

(85) The above examples may be described in connection with a wood-type golf club, the apparatus, methods, and articles of manufacture described herein. Alternatively, the apparatus, methods, and articles of manufacture described herein may be applicable other type of sports equipment such as a hockey stick, a tennis racket, a fishing pole, a ski pole, etc.

(86) Moreover, embodiments and limitations disclosed herein are not dedicated to the public under

the doctrine of dedication if the elements and/or limitations: (1) are not expressly claimed in the claims; and (2) are or are potentially equivalents of express elements and/or limitations in the claims under the doctrine of equivalents.

(87) Various features and advantages of the disclosure are set forth in the following clauses:

(88) Clause 1: A golf club head comprising: a metallic front body including a strike face and a surrounding frame that extends rearward from a perimeter of the strike face; wherein the strike face has a centerpoint, a loft plane tangent to the centerpoint along the strike face, and a midplane extending through the centerpoint from the heel to the toe and perpendicular to the loft plane; a rear body coupled to the metallic front body, wherein the rear body and front body form a substantially hollow structure with a cavity therebetween; the rear body comprises a crown member and a sole member, wherein the sole member is coupled to the crown member, wherein the sole member comprises: a structural layer formed from a filled thermoplastic material, the structural layer including a plurality of apertures extending through a thickness of the structural layer; and a resilient layer bonded to an external surface of the structural layer such that the resilient layer extends across each of the plurality of apertures, wherein the resilient layer is formed from a fiber-reinforced thermoplastic composite material and defines an opening; a metallic weight pad extending at least partially through the opening in the resilient layer and bonded to the structural layer, wherein the metallic weight pad comprises an aperture for the attachment of a metallic weight; and wherein the structural layer and the resilient layer each comprise a common thermoplastic resin component, and wherein the structural layer is directly bonded to the resilient layer without an intermediate adhesive.

(89) Clause 2: The golf club head of clause 1, wherein the metallic front body further includes a flange that is inwardly recessed from an external surface of the surrounding frame; wherein the structural layer of the sole member is adhesively bonded to the flange; and wherein the external surface of the resilient layer of the sole member is flush with the external surface of the surrounding frame.

(90) Clause 3: The golf club head of clause 2, wherein the metallic front body further includes an extension wall that couples the surrounding frame to the flange; wherein the structural layer of the sole member includes a structural member extending towards the metallic front body from the weighted pad; wherein the structural member is operative to transfer a dynamic load between the weight pad and the extension wall during an impact between the strike face and a golf ball.

(91) Clause 4: The golf club head of any of clauses 1-3, comprises a head center of gravity located at a head CG depth from the loft plane, measured in a direction perpendicular to the loft plane, and at a head CG height from the midplane, measured in a direction perpendicular to the midplane; wherein the head CG depth is greater than 1.7 inches.

(92) Clause 5: The golf club head of any of clauses 1-4, wherein the metallic front body further comprises a strike face insert and a receiving frame; wherein the receiving frame has a greater density than the strike face insert.

(93) Clause 6: The golf club head of any of clauses 1-5, wherein the mass of the front body does not exceed 140 g and the total mass of the golf club head does not exceed 210 g.

(94) Clause 7: The golf club head of any of clauses 1-6, wherein a mechanical fastener affixes the metallic weight within the aperture of the metallic weight pad; wherein the aperture of the metallic weight pad of the structural layer comprises threading, and the metallic weight is devoid of threading.

(95) Clause 8: The golf club head of any of clauses 1-7, wherein the metallic weight has a mass ranging from 5 grams to 30 grams.

(96) Clause 9: A golf club head comprising: a metallic front body including a strike face and a surrounding frame that extends rearward from a perimeter of the strike face; wherein the strike face has a centerpoint, a loft plane tangent to the centerpoint along the strike face, and a midplane extending through the centerpoint from the heel to the toe and perpendicular to the loft plane; a rear

body coupled to the metallic front body, wherein the rear body and front body form a substantially hollow structure with a cavity therebetween, the rear body comprises a crown member and a sole member, wherein the sole member coupled to the crown member, wherein the sole member comprises: a structural layer formed from a filled thermoplastic material and bonded to the crown member, the structural layer including a plurality of apertures extending through a thickness of the structural layer; and a resilient layer bonded to an external surface of the structural layer without an intermediate adhesive such that the resilient layer abuts the metallic front body and extends across each of the plurality of apertures; wherein the structural layer is formed from a first material consisting of a first plurality of fibers disposed within a first thermoplastic polymer, and the resilient layer is formed from a second material consisting of a second plurality of fibers disposed within a second thermoplastic polymer, wherein an amount of the first thermoplastic polymer, by volume, within the first material is greater than an amount of the second thermoplastic polymer, by volume, within the second material; wherein the structural layer and the resilient layer each comprise a common thermoplastic resin component, and wherein the structural layer is directly bonded to the resilient layer without an intermediate adhesive; and wherein the structural layer of the sole member includes a metallic weight pad, wherein the metallic weight pad comprises an aperture for the attachment of a metallic weight.

(97) Clause 10: The golf club head of clause 9, wherein the metallic front body further includes a flange that is inwardly recessed from an external surface of the surrounding frame; wherein the structural layer of the sole member is adhesively bonded to the flange; and wherein the external surface of the resilient layer of the sole member is flush with the external surface of the surrounding frame.

(98) Clause 11: The golf club head of any of clauses 9-10, wherein the metallic front body further includes an extension wall that couples the surrounding frame to the bonding flange; wherein the structural layer of the sole member includes a structural member extending towards the metallic front body from the weighted pad; wherein the structural member is operative to transfer a dynamic load between the weight pad and the extension wall during an impact between the strike face and a golf ball.

(99) Clause 12: The golf club head of any of clauses 9-11, wherein the first thermoplastic polymer is directly bonded to the second thermoplastic polymer.

(100) Clause 13: The golf club head of any of clauses 9-12, wherein the first plurality of fibers comprises a plurality of discontinuous fibers, each having a maximum dimension of less than 0.43 inches.

(101) Clause 14: The golf club head of any of clauses 9-13, wherein the second plurality of fibers comprises a plurality of continuous fibers interwoven as a fabric.

(102) Clause 15: The golf club head of any of clauses 9-14, wherein the first thermoplastic polymer is the same as the second thermoplastic polymer.

(103) Clause 16: The golf club head of any of clauses 9-15, wherein the mass of the front body does not exceed 140 g and the total mass of the golf club head does not exceed 210 g.

(104) Clause 17: The golf club head of any of clauses 9-16, comprises a head center of gravity located at a head CG depth from the loft plane, measured in a direction perpendicular to the loft plane, and at a head CG height from the midplane, measured in a direction perpendicular to the midplane; wherein the head CG depth is greater than 1.7 inches.

(105) Clause 18: The golf club head of any of clauses 9-17, wherein the metallic front body further comprises a strike face insert and a receiving frame; wherein the receiving frame has a greater density than the strike face insert.

(106) Clause 19: The golf club head of any of clauses 9-18, wherein a mechanical fastener affixes the metallic weight within the aperture of the metallic weight pad; wherein the aperture of the metallic weight pad of the structural layer comprises threading, and the metallic weight is devoid of threading.

(107) Clause 20: The golf club head of any of clauses 9-19, wherein the metallic weight has a mass ranging from 5 grams to 30 grams.

Claims

1. A golf club head comprising: a heel and a toe opposite the heel; a metallic front body including a strike face and a surrounding frame that extends rearward from a perimeter of the strike face; wherein the strike face has a centerpoint, a loft plane tangent to the centerpoint along the strike face, and a midplane extending through the centerpoint from the heel to the toe and perpendicular to the loft plane; a rear body coupled to the metallic front body, wherein the rear body and the metallic front body form a substantially hollow structure with a cavity therebetween, wherein the rear body comprises a crown member and a sole member, wherein the sole member is coupled to the crown member, and wherein the sole member comprises: a structural layer formed from a molded thermoplastic material, the structural layer including a plurality of apertures extending through a thickness of the structural layer; a resilient layer bonded to an external surface of the structural layer such that the resilient layer extends across each of the plurality of apertures; a metallic weight pad bonded to the structural layer, wherein the metallic weight pad comprises a weight pad aperture for attachment of a metallic weight; and an encapsulating layer between the metallic weight pad and the structural layer of the sole member; a head center of gravity located at a head CG depth from the loft plane, measured in a direction perpendicular to the loft plane, and at a head CG height from the midplane, measured in a direction perpendicular to the midplane; wherein the head CG depth is greater than 1.7 inches; and wherein the crown member comprises a non-metallic material.
2. The golf club head of claim 1, wherein the molded thermoplastic material of the structural layer of the sole member is a filled thermoplastic material.
3. The golf club head of claim 2, wherein the resilient layer of the sole member is formed from a fiber-reinforced thermoplastic composite material.
4. The golf club head of claim 3, wherein the crown member comprises a crown filled thermoplastic layer.
5. The golf club head of claim 4, wherein the crown filled thermoplastic layer comprises a same material as the structural layer of the sole member.
6. The golf club head of claim 1, wherein the crown member comprises a crown filled thermoplastic layer and a crown resilient layer.
7. The golf club head of claim 6, wherein the crown filled thermoplastic layer is formed from a first material consisting of a first plurality of fibers disposed within a first thermoplastic polymer, and the crown resilient layer is formed from a second material consisting of a second plurality of fibers disposed with a second thermoplastic polymer.
8. The golf club head of claim 7, wherein the first thermoplastic polymer and the second thermoplastic polymer share a common thermoplastic resin.
9. The golf club head of claim 7, wherein: the first plurality of fibers comprises a plurality of discontinuous fibers, each having a maximum dimension of less than 0.43 inch; and the second plurality of fibers comprises a plurality of continuous fibers interwoven as a fabric.
10. The golf club head of claim 1, wherein the structural layer of the sole member and the resilient layer of the member sole each comprise a common thermoplastic resin component, and wherein the structural layer is directly bonded to the resilient layer without an intermediate adhesive.
11. The golf club head of claim 1, wherein the crown member comprises a generally curvilinear shape which is concave with respect to a ground plane.
12. The golf club head of claim 1, wherein: the metallic front body further includes a flange that is inwardly recessed from an external surface of the surrounding frame; the structural layer of the sole member is adhesively bonded to the flange; and an external surface of the sole member is flush

with the external surface of the surrounding frame.

13. The golf club head of claim 1, wherein a mass of the metallic front body does not exceed 140 g and a total mass of the golf club head does not exceed 210 g.
 14. The golf club head of claim 1, wherein: a mechanical fastener affixes the metallic weight within the weight pad aperture of the metallic weight pad; the weight pad aperture of the metallic weight pad of the structural layer comprises threading; and the metallic weight is devoid of threading.
 15. The golf club head of claim 1, wherein the metallic weight has a mass ranging from 5 grams to 30 grams.
 16. The golf club head of claim 1, wherein a portion of the metallic weight pad is encapsulated within the structural layer of the sole member, to increase bond strength between the metallic weight pad and the structural layer.
 17. The golf club head of claim 1, wherein: the encapsulating layer is a structural tape that extends over an edge of the metallic weight pad and forms an adhesive bond between the metallic weight pad and the structural layer of the sole member.
 18. The golf club head of claim 1, wherein: the metallic weight pad extends through an opening in the structural layer of the sole member; and the opening is spaced apart from the metallic front body by a minimum distance of at least 25 mm.
 19. The golf club head of claim 18, wherein an internal surface of the metallic weight pad is exposed on an interior of the golf club head.
 20. The golf club head of claim 1, wherein the metallic weight pad comprises a material selected from the group consisting of: stainless steel, titanium, aluminum, and a steel alloy.
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